

TITLE PAGE

- **Problem Statement ID - Ques 1**
- **Problem Statement Title- Eco-Loop**

Building Agents

- **Theme- Software**
- **PS Category- Software**
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EcoJudge - Adversarial Setpoint Court

Proposed Solution (Describe your Idea/Solution/Prototype)

- Two adversarial LLM agents (Energy Advocate, Comfort Advocate) debate HVAC setpoints every control step; a deterministic judge decides using EnergyPlus-measured comfort (PMV), not a vote or an LLM guess.
- Static rule-based BMS schedules cannot adapt to real-time conditions. This closed loop continuously re-simulates the building and adjusts setpoints using live measured feedback from EnergyPlus.
- Combines adversarial multi-agent arbitration, an LLM self-repair loop for failed patches, and a precedent cache that cut repeat LLM calls by 70 percent in testing – all grounded in real physics simulation, not just LLM

TECHNICAL APPROACH

- EnergyPlus 23.2 (physics simulation), eppy (building model editing), Python, open-source Llama 3.3 70B via Groq API, MCP tool server, pytest (27 automated tests).
- EnergyPlus measures zone temperature, humidity, energy use and PMV. Energy and Comfort agents each propose setpoints. A deterministic judge picks the winner using measured PMV. Winning setpoints are patched into EnergyPlus and it re-simulates. A self-repair loop retries automatically if a patch causes a simulation error.

FEASIBILITY AND VIABILITY

- Fully working prototype: 27 of 27 automated tests passing, verified end-to-end on two independent machines, with a real 3-day EnergyPlus run producing measured results (10.1 percent energy reduction vs baseline).
- LLM API latency and cost; dependency on a local EnergyPlus install; risk of simulation errors from bad setpoints.
- Precedent cache cuts repeat LLM calls by 70 percent; self-correction loop auto-repairs failed patches using

ARTIFACTS

- Relevant artifacts, such as :
 - Full source code: github.com/Ridanshi/EcoJudge
 - Demo video and full agent-debate transcript included in the repository
 - Dashboard: measured energy and comfort, baseline vs closed-loop (see results/dashboard.png)

RESEARCH AND REFERENCES

- EnergyPlus and eppy documentation; ISO 7730 PMV thermal comfort standard; Model Context Protocol (MCP) specification. Repository: github.com/Ridanshi/EcoJudge